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Predicting Internal Migration in the Brazilian Amazon

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INTRODUCTION

The Brazilian Amazon is experiencing deforestation, rising temperatures, and population loss—often in the same places. We set out to examine whether it is possible to predict where people move next by integrating satellite, census, and climate data.

Rather than asking how many people leave a given municipality, we model migration as flows between **origin–destination pairs (dyads)**. There are 5,570 municipalities in Brazil and 300K+ pairs exhibiting non-zero flows. This framework allows us to capture not just the scale of out-migration, but the *direction* of movement, an essential dimension for understanding population dynamics in the Amazon.

~17%
of the Amazon's original forest is gone

+0.5°C
Brazil has warmed since 1960

5,570
municipalities in the model

300K+
town-to-town flows

OUR DATA

We integrate four complementary datasets, linked using IBGE municipality codes and spanning 2005–2025. Each captures a distinct dimension of local conditions that are often studied in isolation:

Satellite — Copernicus NDVI

A monthly vegetation index used to detect changes in greenness, serving as a key proxy for **deforestation**.

Vegetation · Deforestation

Census — IBGE 2010

Provides demographic and socioeconomic characteristics, offering a snapshot of who lives where and under what conditions.

Income · Education · Electricity · Sewage

Climate — ERA5

Measures temperature and precipitation anomalies, capturing deviations from typical weather patterns.

Temp · Precip · Humidity · UV

Agriculture — IBGE SIDRA

Details agricultural activity, including crop types and the extent to which populations rely on farming.

KEY FEATURES

Before building the model, we examined the basic patterns: **who is actually leaving?** We grouped municipalities into quintiles based on temperature, income, and electricity access, and then compared the average net migration rate across these groups.

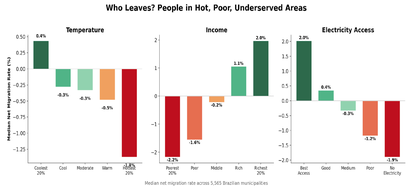


Figure 1. Median net migration rate by temperature, income, and electricity-access quintile (n = 5,565).

The patterns are clear and consistent. Municipalities in the **hottest** quintile lose population roughly 3× faster than those in the coolest. The **poorest** quintile experiences a decline of about 2.2%; the richest sees gains of 2.0%. Areas lacking electricity show the highest rates of out-migration across all infrastructure categories.

These relationships are smooth and directionally consistent, so we retained them as model features—while treating them as descriptive patterns rather than causal conclusions.

Features retained

- **Agricultural employment** — capturing the farm belt along the Amazon arc.
- **Infrastructure gaps** — including lack of electricity and sewage access.
- **Vegetation loss (NDVI)** — a proxy for deforestation.
- **Education & income** — key factors influencing destination choice.

> These results reflect **correlations, not causal mechanisms**. Determining whether factors such as heat or poverty directly drive migration remains an important question for future work.

By the numbers

~3×
hottest vs. coolest outflow gap

–2.2%
net migration, poorest fifth

+2.0%
net migration, richest fifth

OBSERVED PATTERNS

Column 2 identifies **which** municipalities are losing people. The next step is to place those patterns in geographic context to inform our modeling choice.

1. **Migration is about pairs, not individual locations.** Migration is inherently directional: people move from one municipality to another. A relatively small number of urban centers absorb flows from many rural origins. Goiânia and Cuiabá draw migrants from broad surrounding areas.

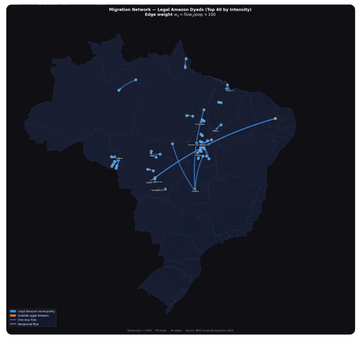
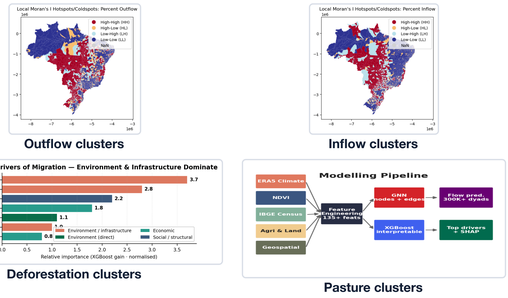


Figure 2. Top-40 origin–destination pairs in the Legal Amazon, weighted by $\texttt{flow}_{ij}/\texttt{pop}_i$. Goiânia and Cuiabá pull flows from a wide rural catchment.

2. **Neighboring municipalities exhibit spatial similarity.** Local Moran's I reveals strong clustering in both migration outcomes and their underlying drivers. Deforestation clusters along the Amazon arc, pasture expansion runs through the cerrado, and areas of high out-migration overlap with both. In contrast, in-migration clusters are concentrated in major urban areas in Brazil's Southeast.

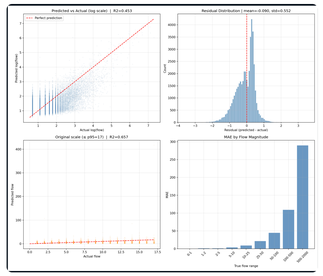


`Figure 3.` Red = high–high clusters, blue = low–low. The **deforestation** cluster sits along the Amazon arc, **pasture** runs through the cerrado, and **outflow** overlaps both. Inflow clusters sit in the Southeast cities.

3. **Implications for modeling.** The signal lies in **spatial relationships and origin–destination pairs**, not in isolated municipalities. Models that treat each unit as independent risk missing these structural dependencies.

MODELLING APPROACH

Figures 2 and 3 define the core requirements for the model: it must capture the origin–destination structure of migration while remaining interpretable. To do this, we combine a **graph neural network (GNN)** to model spatial and network structure with **XGBoost + SHAP** for feature-level explanation.



PRELIMINARY **Figure 4.** Held-out GNN fit. **R^2 = 0.453** on log-flow, **R^2 = 0.657** on the original scale (≤ p95). Residuals: mean –0.09, std 0.55. Error is bigger on the largest flows — that's where the model struggles most.

Current status. The GNN is still training. Preliminary results show:

- **R^2 = 0.453** on log-transformed flows; **R^2 = 0.657** on the original scale (≤ p95).
- **Residuals:** mean –0.09, std 0.55 — indicating slight underprediction on average.
- **Error increases with flow size**, with the largest flows posing the greatest challenge.

WHY A GNN — capturing spatial structure

- Municipalities are modeled as **nodes** and migration flows as **edges**, matching the dyadic structure shown in Fig. 2.
- Message passing allows information to propagate across neighboring municipalities.
- Architecture: **GraphSAGE + attention**, 2 layers, 64-d hidden, trained using MSE on log-transformed flows.

WHY TREE-BASED MODELS — enabling interpretability

- **XGBoost** (500 trees, depth 6) with 5-fold **spatial cross-validation**, where folds are defined as large contiguous regions to avoid spatial leakage.
- **SHAP** values identify which of the 135 features contribute most to predictions.
- The GNN is used for prediction, while the tree-based model provides interpretability.
- To justify the added complexity of the GNN, it must outperform XGBoost under the same spatial cross-validation scheme. This head-to-head comparison is still in progress — the GNN predicts; the tree explains.

IMPACT — AMAZON

Applying the trained GNN across all municipalities in the Legal Amazon can translate these patterns into actionable insights:

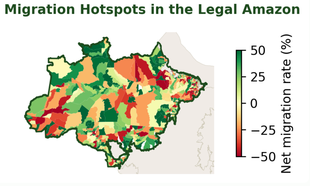


Figure 6. Model-scored net migration rate across the Legal Amazon. Red = losing population.

- A **ranked list of at-risk Amazon towns** — useful for directing climate-adaptation funding.
- A **ranked set of at-risk municipalities** — supporting more targeted allocation of climate adaptation resources.
- **Policy alignment** — informing programs such as **PPCDAm**, **Bolsa Verde**, and **REDD+** by linking deforestation hotspots to observed population displacement.

FUTURE WORK — AMAZON

Our next steps focus on incorporating more recent data and moving toward causal inference:

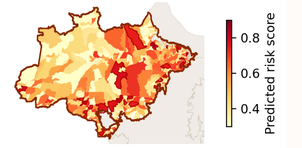


Figure 7. Predicted risk score — priority Amazon towns for 2022 validation.

- **Update with new census data** — integrate the 2022 Census to capture post-pandemic migration dynamics in the Amazon.
- **Advance causal analysis** — move beyond correlation (double-ML, synthetic controls), centered on specific shocks (e.g., the 2019 Amazon fire season).
- **External validation** — compare the model's priority map with 2022 displacement records in the Amazon region.

Data: IBGE Censo Demográfico, Copernicus NDVI, ERA5, IBGE SIDRA · We thank NCSA for computational support and Prof. Lyons, Prof. Soliman & D. Pingali for mentorship.